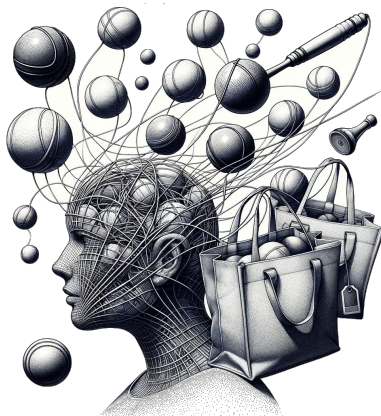




HypergraphAI

Semantic Knowledge for AI Era

One store of meaning, provenance and time:
for people,
for AI agents,
and for every model vendor they use.



The Knowledge Representation Bottleneck

Limitations of Legacy Data Structures

- **Relational Tables:** Make complex relationships purely implicit.
- **Property Graphs:** Force multi-entity real events into artificial pairs.
- **Vector Stores:** Lack structure, provenance, and temporal context.

Impact on AI Agents

Frontier models reason well, but because enterprise knowledge is stored in structures built for other eras, AI agents **hallucinate relationships, forget across sessions, and cannot be audited.**

Gartner expects more than 40% of agentic AI projects to be cancelled by the end of 2027, citing cost, unclear value, and inadequate risk controls like grounding and provenance.

— Gartner Market Analysis

Solution Overview

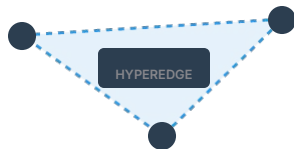


01 / DATA MODEL

First-Class N-Ary Hyperedges

HypergraphAI models real-world complexity using first-class n-ary hyperedges that join any number of nodes seamlessly.

- Attributes & Custom Tags
- Native Validity Windows
- Complete Audit Trails



Joins N Nodes Direct

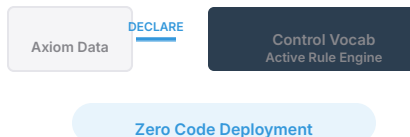
02 / SEMANTIC CONTROL

Data-Driven Semantic Rules

Relation meaning is not hard-coded; it is declared as axiom hyperedges using a controlled vocabulary.

Data Change vs Code Deployment

Adding an inference rule is purely a data change, requiring zero code deployment.

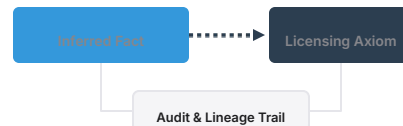


03 / EXPLAINABILITY

Live Explainable Inference

Inferred facts are computed live at query time, flagged as inferred, and point directly to the licensing axiom.

- Real-Time Query Computing
- Explicit Inferred Flags
- 100% Explainable Provenance





System Architecture

HypergraphAI combines knowledge-graph semantics with document-database flexibility and an AI-native interface.

Data Model Flexibility

- Attributes are free-form documents
- Schema migrations are never required
- Ontology itself is treated as data

~26k

Lines of Code

346

Automated Tests

INTERFACE LAYER

AI-Native Interface & REST / FastAPI

Exposes seamless semantic query boundaries and structured RESTful endpoints for enterprise interaction.

CORE ENGINE

Hybrid Knowledge Graph & Document Engine

MIT-Licensed core built on Python 3.11+

High-performance, strongly typed hypergraph modeling combining entity graph paths with flexible document payload structures.

STORAGE LAYER

Pluggable Storage Abstraction

Default MongoDB 7 backend with seamless modular persistence adapters for enterprise backends.

SHQL & Deterministic Execution Engine



QUERY LANGUAGE

SHQL Query Language

A SPARQL-inspired, YAML-native pattern language supporting multi-hop traversal, implicit joins, and complex aggregation.

Because it is natively structured in YAML, LLMs can reliably generate, parse, and repair queries without syntax errors or loss of semantics.

TRANSPARENT REASONING

Live Explainable Inference

Inference executes dynamically at query time and is never silently persisted to the underlying store.

Every derived edge explicitly declares its exact origin source and underlying mathematical axiom for 100% explainable and audit-ready reasoning paths.

TEMPORAL AUDITING

Point-in-Time Queries

Every hypernode and hyperedge maintains discrete, explicit validity windows across its entire lifecycle.

Queries support an explicit 'at' parameter to instantaneously reconstruct and retrieve the exact state of the world at any historical second.



Native MCP Infrastructure & HgNexus

30 Native

Model Context Protocol (MCP) Tools

HypergraphAI features 30 native MCP tools that allow any MCP-capable model to execute platform operations using the caller's permissions.

This vendor-neutral infrastructure plugs directly into models like **Claude** and **ChatGPT** without requiring custom connectors.

HgNexus Agent Interface

A built-in, multi-vendor agent chat interface native to the platform.

HgNexus stores multi-turn context server-side, persists prompt history, and seamlessly leverages the platform's own MCP tools.

Bring Your Own Key (BYOK)

Complete security and multi-model flexibility.

Users can bring their own encrypted API keys to interact with multiple foundation models directly within the HypergraphAI environment.



Enterprise System of Record & Semantic Layer

GOVERNED STORE

System of Record for Meaning

- **Unified Data Store:** Unifies entities, n-ary relationships, ontology, provenance, and history into one governed store.
- **Enterprise Governance:** Features spaces and role-based access control (RBAC) to cleanly separate business units.
- **Complete Auditability:** Full historical tracking and lineage preservation across all semantic relationships.

FEDERATED ARCHITECTURE

Shared Semantic Layer

- **Ontology as Data:** Acts as a shared semantic layer across existing systems by providing a unified shared ontology.
- **Mesh Federation:** Enables federated queries across multiple servers without physically moving underlying datasets.
- **Unified Agent Vocabulary:** Allows every system and AI agent to utilize one shared vocabulary over data that remains exactly where it lives.



Vendor-Neutral Memory Layer

Context belongs entirely to the customer, rather than model vendors or chat interfaces.

01 / SOVEREIGNTY

Customer-Owned Memory

- Decoupled entirely from LLM vendors and chat product interfaces.
- Memory layer acts as dedicated infrastructure owned entirely by the customer.
- Eliminates vendor lock-in at both the memory and context layer.

02 / PERSISTENCE

Multi-Session & Cross-Model

- Spans across different model vendors and evolving architectures.
- Accessible by any MCP or REST client today or in the future.
- Persists across sessions, ensuring memory outlives any individual chat.

03 / PORTABILITY

Portable & Governed Context

- Mesh federation enables sharing across diverse hosting environments.
- Seamless one-file export and import capabilities for complete mobility.
- Governed strictly by granular, space-level enterprise access controls.



Scale & Data Fidelity: Alchemy Fraud Graph

3.63M

Hypernodes Generated

1.87M

Hyperedges Created

100%

Provenance Metadata

Alchemy Platform Demonstration

HypergraphAI processed six CSV files from disparate sources (data modelled differently between sources) containing 2,000,000 transactions into 3,631,632 hypernodes and 1,872,417 hyperedges, complete with a generated domain ontology, in approximately 13 minutes.

Crucially, 100% of these nodes and edges carry complete provenance metadata, with zero dangling references across 13.5 million hyperedge member slots. A random sample of 2,000 source rows compared against stored data revealed zero mismatches, proving the platform's ability to maintain exact data fidelity at scale.



Aggregate Market Opportunity Highlights

NET TAM (2030)

\$28.0B

24 de-duplicated use cases

NET SAM

\$1.2B

Target orgs & reachability focus

YEAR-5 SOM (ARR)

\$37M

Execution-constrained base

KEY MARKET TAKEAWAYS

- The aggregate net TAM is estimated at **\$28.0B** in 2030, carefully de-duplicated across 24 use cases.
- The net Serviceable Available Market (SAM) reaches **\$1.2B**, focusing on target organizations and reachability.
- The execution-constrained Year-5 Serviceable Obtainable Market (SOM) is **\$37M ARR**.
- Value is heavily concentrated in the agentic core, which alone contributes **\$29M** of the base SOM.

Standardized Methodology for Market Sizing



- **Total Addressable Market (TAM)** reflects 2030 annual spend, derived from analyst reports or bottom-up population models.
- **Serviceable Available Market (SAM)** is strictly bottom-up, multiplying target organizations by need share, reachability, and ACV.
- **Serviceable Obtainable Market (SOM)** represents execution-constrained Year-5 ARR.
- To ensure accuracy, aggregate figures apply a **0.85 cross-cluster retention factor** to remove shared account overlaps and prevent double counting.

The Agentic Core Wedge Drives Primary Value



USE CASE 6

AI-Agent Persisted Context Stores

First half of the core wedge, establishing baseline memory infrastructure for autonomous AI execution workflows.

USE CASE 7

Agentic Semantic Knowledge Stores

Second half of the core wedge, providing structured semantic reasoning and deep contextual knowledge retrieval.

COMBINED MARKET OPPORTUNITY

\$486M SAM / **\$29M**
SOM

Addressable and serviceable market generated jointly by Use Case 6 and Use Case 7.

AGGREGATE SOM SHARE

79%

Proportion of the execution-constrained aggregate SOM captured by this agentic core.

These core markets closely reproduce the investor deck's original assumptions and form the foundation of the funded seed plan.



Expansion into Enterprise Horizontal Spaces

Beyond the core wedge, horizontal enterprise spaces offer significant upside.

Market Segment 01

Enterprise Semantic Knowledge Stores

TAM

\$5.7B

SOM

\$3.4M

Market Segment 02

Enterprise Semantic Integration Layer

TAM

\$4.6B

SOM

\$2.2M

Market Segment 03

Enterprise Relationship Analytics

Driven by the need for multi-hop and temporal analysis alongside conventional BI.

TAM

\$5.4B

SOM

\$1.6M

Vertical Solutions Offer Partner-Led Growth



Vertical solutions represent adjacent, Tier 2 and Tier 3 opportunities best pursued via partners.

Tier 2 Opportunity

Fraud & AML Analytics

\$3.0B TAM & **\$1.3M** SOM

Fraud and AML analytics present a \$3.0B TAM and a \$1.3M SOM.

Tier 2 Opportunity

Defense & Intelligence Analysis

\$1.2B TAM & **\$0.81M** SOM

Defense and intelligence analysis offers a \$1.2B TAM and a \$0.81M SOM.

Tier 3 Expansion Pathway

Software Engineering Architecture

\$1.47B Combined TAM

Software engineering architecture and GRC knowledge graphs contribute an additional \$1.47B in combined TAM, providing long-term strategic expansion pathways.

Tier 3 Expansion Pathway

GRC Knowledge Graphs

\$1.47B Combined TAM

Software engineering architecture and GRC knowledge graphs contribute an additional \$1.47B in combined TAM, providing long-term strategic expansion pathways.

Strategic Upside in First-of-Kind Markets



MARKET OVERVIEW: Emerging spaces create category defensibility with a combined base **TAM of \$4.7B** and **SOM of \$6.2M**.

TARGET SOM

\$2.2M

Multi-Agent Shared World Models

Multi-agent shared world models represent a \$2.2M SOM by governing temporal state for agent fleets.

TARGET SOM

\$1.5M

Agent Accountability Ledgers

Agent accountability ledgers offer a \$1.5M SOM for AI audit evidence.

TARGET SOM

\$0.90M

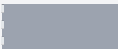
Portable, User-Owned AI Memory

Portable, user-owned AI memory offers a strong first-of-kind claim, addressing vendor-neutral memory portability with a \$0.90M SOM.

Revenue Concentration in Top Use Cases



The obtainable market is highly concentrated among a few critical use cases.

USE CASE APPLICATION	SERVICEABLE OBTAINABLE MARKET (SOM)	SOM VALUE
Agentic Semantic Knowledge Store		\$16.0M
AI-Agent Persisted Context		\$13.0M
Enterprise Semantic Knowledge		\$3.4M
General-Purpose Graph Stores		\$2.9M
Multi-Agent Shared World Models		\$2.2M

Reconciling the Aggregate Model with the Investor Deck



Baseline Strategy

Core Funded Plan

Core Wedge SAM

\$486M

Core Wedge SOM

\$29M

Key Takeaways:

- The original investor deck remains **completely valid** as the core-wedge case.
- The core wedge SAM of **\$486M** and SOM of **\$29M** perfectly mirror the baseline deck's assumptions.
- If presented together, the **\$29M SOM** represents the fully funded operational plan.

Expansion Strategy

Unfunded Upside & Expanded Scope

Net TAM

\$28.0B

Expanded SOM

\$37M

Reconciliation Delta

+\$8M SOM represents unfunded upside bringing total SOM from \$29M to \$37M.

- The expanded **\$37M SOM** and **\$28.0B Net TAM** incorporate adjacent horizontal and vertical market spaces.
- Captures additional value potential without altering baseline unit economics or core execution risk.



Scenario Analysis Highlights Robust Upside

Market sizing utilizes three scenarios based on analyst spreads and obtainable share inputs:

Conservative Yields a **\$12M SOM** and **\$769M SAM**

Base Projects a **\$37M SOM** and **\$1.2B SAM**

Optimistic Forecasts an **\$80M SOM** and **\$1.8B SAM**

Sensitivity Factor

Outcomes are highly sensitive to enterprise agent adoption rates and obtainable share metrics.



Execution Tiering Mitigates Operational Risk

Capacity Limitation Context: The total unconstrained SOM is **\$58M**, but it is limited by execution capacity to **\$37M**.

TIER 1 1.0 FACTOR

Core Wedge Use Cases

Tier 1 core wedge use cases are pursued at a 1.0 execution factor.

TIER 2 0.5 FACTOR

Adjacent Spaces

Tier 2 adjacent spaces are discounted to a 0.5 execution factor, requiring packs or partners.

TIER 3 0.2 FACTOR

Opportunistic Spaces

Tier 3 opportunistic spaces are heavily discounted to a 0.2 execution factor.

This tiering framework actively manages the opportunity cost of focus.

Investment Conclusion and Strategic Growth



Net TAM

\$28.0B

Total net addressable market opportunity

Net SAM

\$1.2B

Adjacent enterprise & vertical expansion

Year-5 SOM

\$37M

Seed plan funds \$29M core wedge

Scalable Expansion Strategy

Expanding into the \$1.2B net SAM through adjacent enterprise and vertical spaces provides a clear, scalable growth trajectory.

Long-Term Defensibility

First-of-kind spaces establish long-term category defensibility for subsequent funding rounds.

HypergraphAI presents a compelling \$37M Year-5 SOM opportunity within a \$28.0B Net TAM.

The seed plan fully funds the highly concentrated \$29M agentic core wedge.

Competitive Architecture Wedge

Governed enterprise semantic knowledge meets agent-native delivery.



Legacy Infrastructure

HypergraphAI Advantage

Property Graphs

Restricted to rigid binary edges that fail to natively capture complex multi-entity enterprise facts.

✓ HypergraphAI N-Ary Objects

Treats n-ary facts as first-class objects out of the box, preserving multi-dimensional relationships without relational fragmentation.

Unstructured Vector Stores

Opaque, probabilistic similarity search lacking deterministic reasoning, enterprise auditability, or temporal awareness.

✓ Explainable Axiom Inference & Point-in-Time Querying

Provides deterministic, axiom-driven inference alongside 100% provenance and historical temporal querying for strict auditability.

Pure Agent-Memory Layers & Siloed DBs

Lacks foundational enterprise grounding, leading to state drift, high operational complexity, and vendor lock-in.

✓ Open-Core Grounded MCP Infrastructure

Delivers native Model Context Protocol (MCP) integration, combining operational simplicity with an extensible open-core foundation.

Strategic Value: HypergraphAI delivers the structural grounding and operational simplicity that legacy databases and ungrounded agent layers lack.



Open-Core Monetization Strategy

Keeping the MIT-licensed core free to drive adoption across seven distinct, high-margin revenue streams.

Primary Recurring Stream

1. Managed Hosting & Enterprise Cloud

Fully managed cloud infrastructure and dedicated enterprise subscriptions, offering automated scaling, high availability, and turn-key operational management.

Ecosystem Expansion

2. Extension Module Marketplace

Monetized hub for specialized add-ons, enterprise integrations, and domain-specific graph algorithms developed by HypergraphAI and ecosystem partners.

Enterprise & Future

3. Support SLAs, Services & Future Streams

- Tiered support SLAs and professional services for ontology design and agent integration.
- Future pipeline: Training certifications and knowledge brokering for curated domain graphs.

Margin Protection

4. Pass-Through Token Economics

Zero AI Resale Risk

AI vendor model costs are strictly pass-through. Customers bring their own API keys, completely insulating software margins from underlying LLM usage costs.



Go-To-Market Strategy

Our go-to-market strategy executes across three motions: **product-led open-source adoption**, **direct commercial sales for managed hosting**, and a **partner channel**.

MONTHS 1-3

MOTION: PRODUCT-LED OS

Core OS Launch

Launch of our open-source core and initial commercial module within the first three months to drive community adoption.

MONTHS 5-9

MOTION: DIRECT COMMERCIAL

Alpha Phase

Onboard 1 to 3 design-partner hosting customers in our Alpha phase to validate managed hosting workflows.

MONTHS 9-15

MOTION: COMMERCIAL GA & PARTNERS

Commercial GA

General availability for commercial hosting featuring public pricing and self-serve options, paving the way for Series A readiness.



Seed Investment & Allocation

\$3.0M

Seed Investment Ask

17 Mo.

Runway to Series A

Capital Allocation

54% Engineering Allocation

Dedicated to building core technical team

Deal Structure

\$11.0M Pre-Money Valuation

Value Creation & Target Milestones



Month 3

Open-Source Launch

Release core open-source hypergraph engine to establish market presence and developer community.



Month 5

Alpha Customers

Secure initial design partners and enterprise alpha customers to validate product-market fit.



Month 9

Commercial GA

Achieve full General Availability for enterprise platform with commercial SLA and deployment tools.



Month 9-17

Series A Raise

Initiate Series A fundraising with key operational metrics, commercial traction, and GA platform.